

Advanced Lesson 9 - Exercises: Defaults, Definition Order, Tags and Attributes

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MATH 7013 - Advanced Mathematical Modeling

1: Optional arguments with default values

```
Clear[f, x, y, z];  
f[x_, y_ : 10, z_ : 20] := {x, y, z}  
?  
{f[1, 2, 3, 4], f[1, 2, 3], f[1, 2], f[1], f[]}
```

2: Definition order - anything, odd, even, and the literal 2

```
Clear[f, i];  
f[x_] := "x is anything";  
f[x_?OddQ] := "x is odd";  
f[x_?EvenQ] := "x is even";  
f[2] := "x = 2"  
?  
TableForm[Table[{i, f[i]}, {i, 1, 4,  $\frac{1}{2}}$ }]
```

```

Clear[f, i];
f[x_?OddQ] := "x is odd";
f[x_] := "x is anything";
f[x_?EvenQ] := "x is even";
f[2] := "x = 2"

?f

TableForm[Table[{i, f[i]}, {i, 1, 4,  $\frac{1}{2}}$ }]

Clear[f, i];
f[2] := "x = 2";
f[x_?OddQ] := "x is odd";
f[x_?EvenQ] := "x is even";
f[x_] := "x is anything"

TableForm[Table[{i, f[i]}, {i, 1, 4,  $\frac{1}{2}}$ }]

```

3: Four tests and a catch-all, and why the order changes the answers

```

Clear[f, i];
f[x_?OddQ] := " x is odd";
f[x_?Positive] := " x is positive";
f[x_?IntegerQ] := " x is an integer";
f[2] := " x = 2";
f[x_] := " x is arbitrary"

?f

TableForm[Table[{i, f[i]}, {i, -2, 4,  $\frac{1}{2}}$ }]

Clear[f, i];
f[x_?OddQ] := " x is odd";
f[x_?IntegerQ] := " x is an integer";
f[x_?Positive] := " x is positive";
f[2] := " x = 2";
f[x_] := " x is arbitrary"

TableForm[Table[{i, f[i]}, {i, -2, 4,  $\frac{1}{2}}$ }]

```

4: Teaching Integrate about a new function, by tagging

```
ClearAll[myFunction, myFunctionIntegral, x, y];
 $\int \text{myFunction}[x_] \, dx := \text{myFunctionIntegral}[x]$ 
ClearAll[myFunction];
myFunction /:  $\int \text{myFunction}[x_] \, dx := \text{myFunctionIntegral}[x]$ ;
??myFunction
 $\int \text{Sin}[x] \, dx$ 
 $\int \text{myFunction}[x] \, dx$ 
 $\int \text{myFunction}[y] \, dy$ 
Head[ $\int \text{myFunction}[x] \, dx$ ]
```

5: An orderless function

```
ClearAll[myFunction, a, b, c, d];
SetAttributes[myFunction, Orderless];
??myFunction
{myFunction[1, 2, 3], myFunction[3, 2, 1]}
{myFunction[a, b, c, d], myFunction[d, c, b, a]}
myFunction[c, b, a, d] == myFunction[d, c, b, a]
```

6: Holding only the first argument

```
ClearAll[myFunction, a, b, c, d];
SetAttributes[myFunction, HoldFirst];
??myFunction
a = 1;
b = 2;
c = 3;
d = 4;
myFunction[a, b, c, d]
myFunction[a_, b_, c_, d_] := {a, b, c, d}
```

```
??myFunction
```

```
a = 1;
```

```
b = 2;
```

```
c = 3;
```

```
d = 4;
```

```
myFunction[a, b, c, d]
```

7: A usage message and full holding

```
ClearAll[myFunction, x, y, z];
```

```
myFunction::usage = "myFunction [x, y, z, ... ] is important to understand.";
```

```
myFunction[x_, y_, z_] := Sin[x y z];
```

```
SetAttributes[myFunction, HoldAll];
```

```
?myFunction
```

```
??myFunction
```